Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

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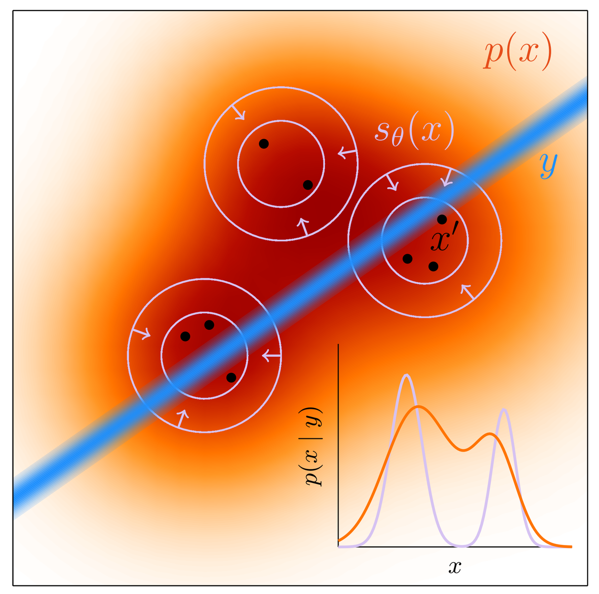
The question

A trained diffusion model can produce mock HI maps in seconds instead of CPU-hours. That speed is only useful if the maps are genuinely new — a model that memorizes its training set gives a biased prior. We train diffusion models on CAMELS HI maps and ask:

1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?



Toy picture: a memorized prior is a set of islands around the training data. The posterior gets pulled toward those islands even when the likelihood is sharp.

Data & models

- **CAMELS** (IllustrisTNG) LH set: 1000 simulations → 128 slices each → 128,000 candidate 128×128 HI maps; 32 held out.
- Six-parameter conditioning: $\boldsymbol{\theta} = (\Omega_m, \sigma_8, A_{\rm SN1}, A_{\rm AGN1}, A_{\rm SN2}, A_{\rm AGN2})$.
- **Diffusion objective.** The model learns to undo noise: corrupt a real map, train a UNet to predict the velocity target at every noise level; the cosmology enters as conditioning.

$$x_t = \sqrt{\bar\alpha_t}\, x_0 + \sqrt{1-\bar\alpha_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$
$$\mathcal{L}_{\rm diff} = \mathbb{E}_{x_0,\epsilon,t}\, \| f_\theta(x_t, t, \boldsymbol{\theta}) - \text{target} \|_2^2 .$$

Here the config uses `v_prediction`: target $= v_t = \sqrt{\bar\alpha_t}\epsilon - \sqrt{1-\bar\alpha_t}x_0$. Ho et al. 2020 (DDPM).

- **Sampling**: 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep**: $N_{2D} = 2^6 \to 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

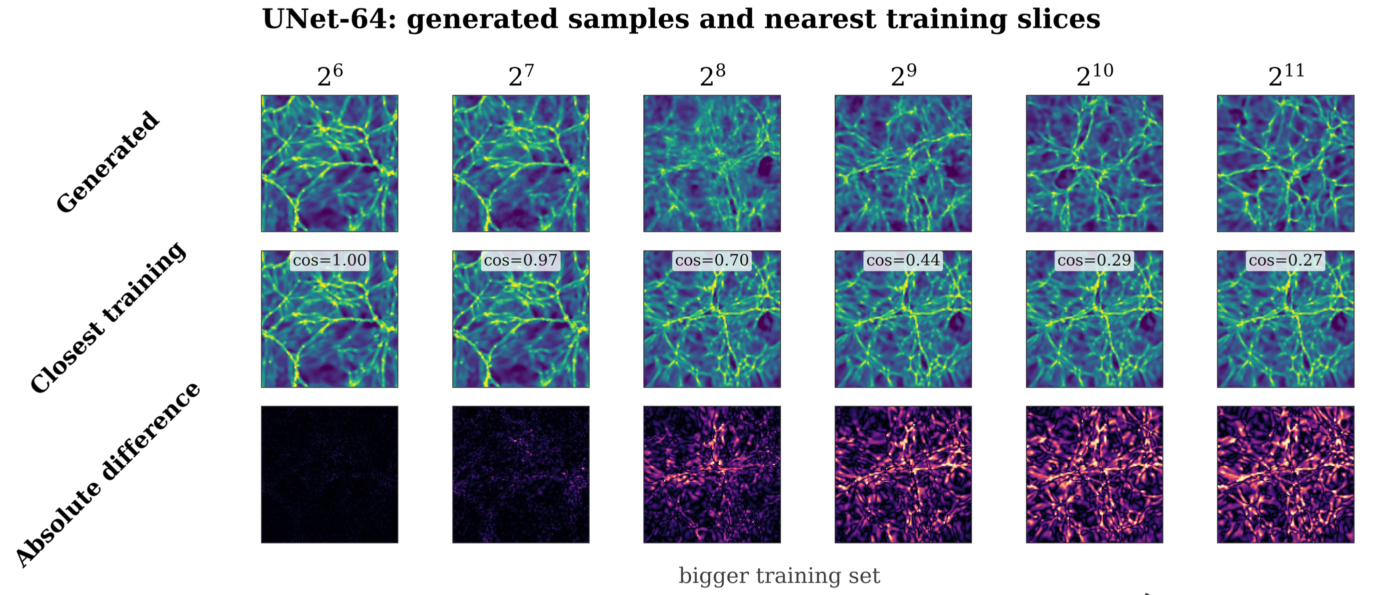
Detecting memorization

For each generated sample x_j, find its nearest training field in pixel, PCA, and SSCD embedding space:

$$s_j = \max_i \, \mathrm{sim}(x_j, y_i), \qquad \mathrm{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

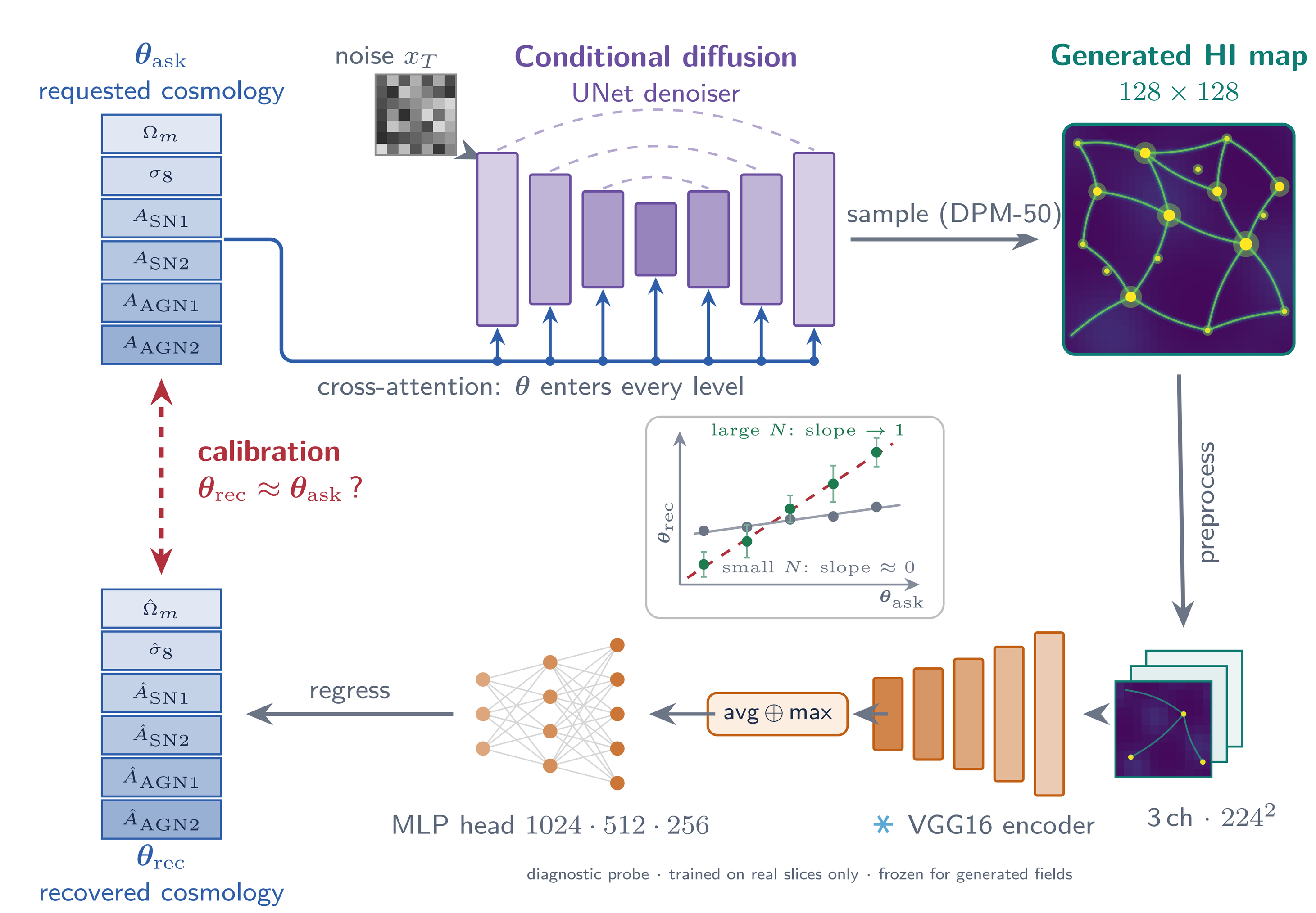
Here τ is set from train-vs-train nearest-neighbor similarity (95th percentile). Independent models should reproduce the same distribution, not the same training images.

UNet-64: generated samples and nearest training slices



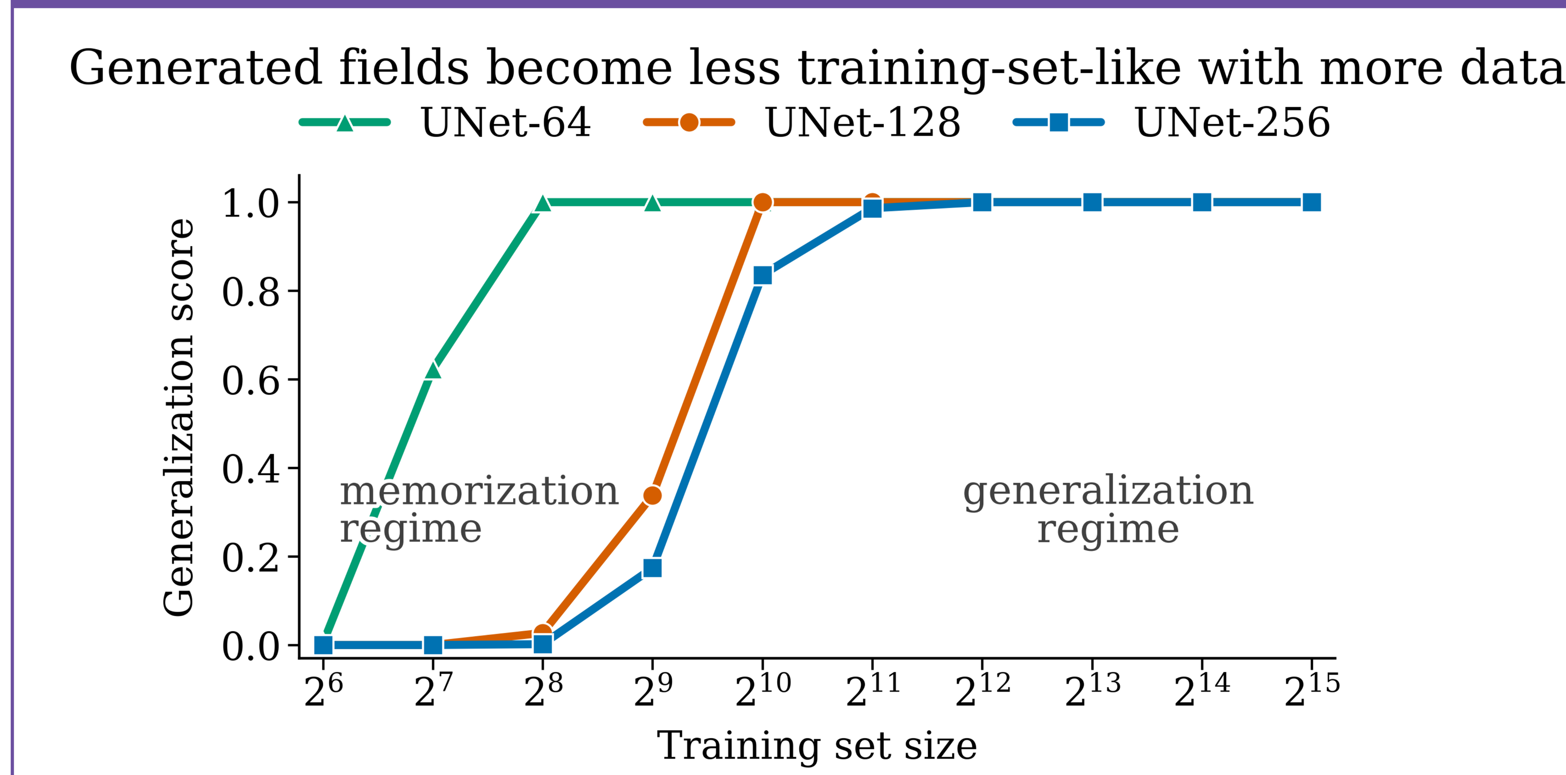
Calibration: does the generator obey θ?

Ask the conditional model for **held-out samples**, then read the cosmology back from its generated maps with a **frozen encoder**. If the conditioning is used, changing $\boldsymbol{\theta}_{\rm ask}$ should move the recovered $\boldsymbol{\theta}_{\rm rec}$ rather than leaving it near the training-set average.



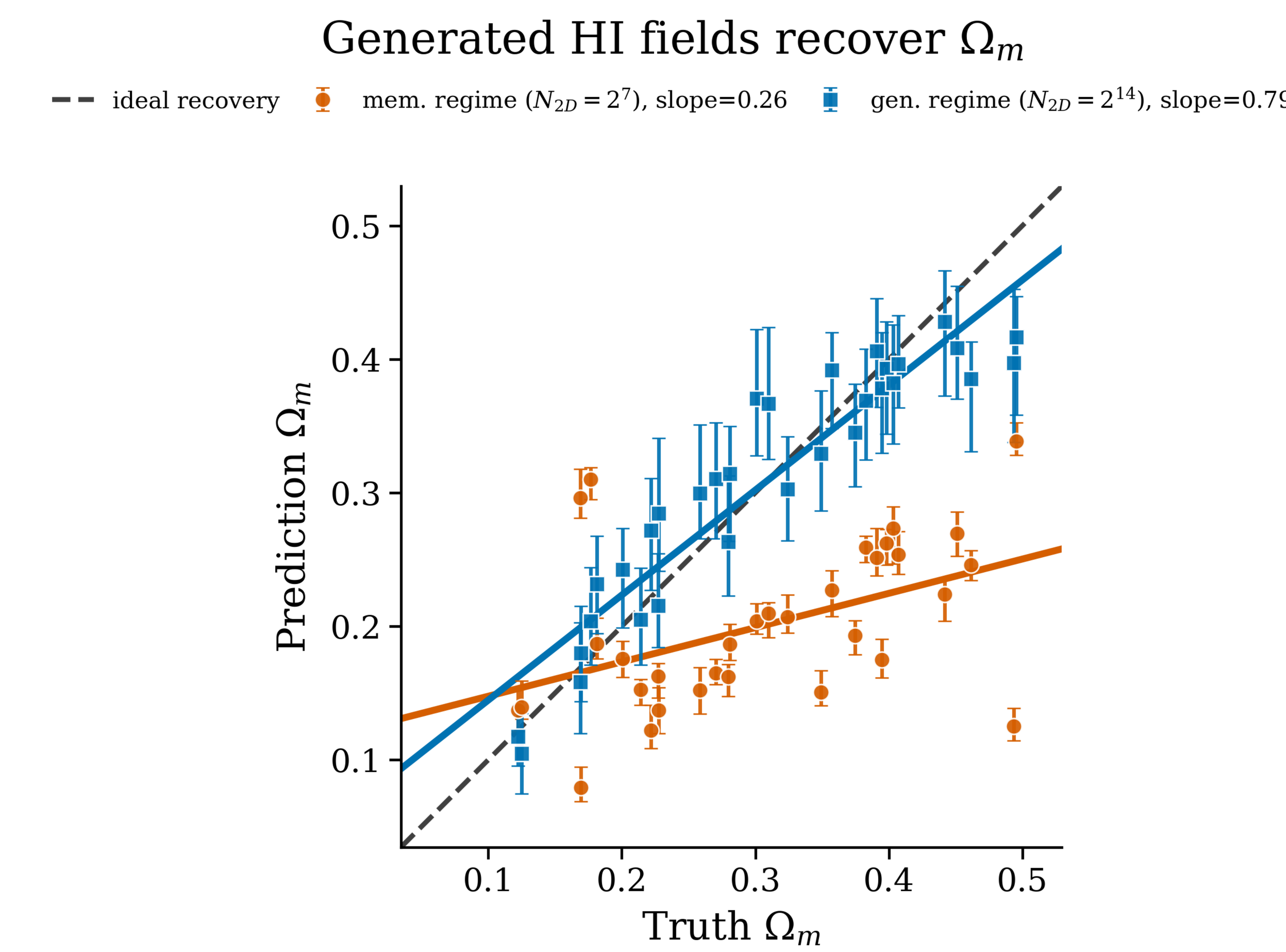
Encoder setup: VGG16 is frozen (ImageNet); only the regression head is trained on *real* slices from non-held-out simulations. Held-out test simulations are used only for encoder validation and generator calibration.

Memorization → generalization transition



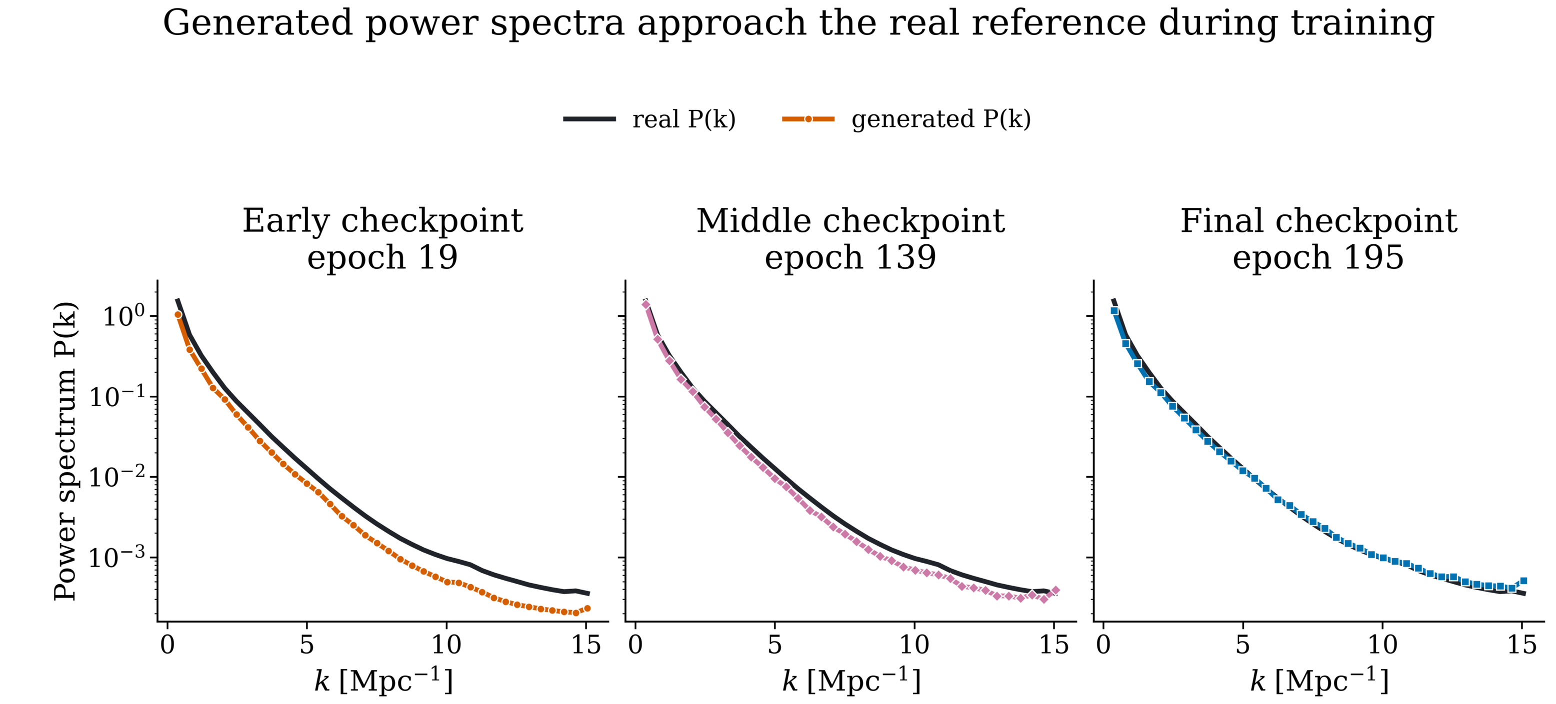
- Score uses a **q95-calibrated threshold** (τ = 95th pct train–train NN similarity): higher means generated fields are less training-set-like in PCA space.
- **Small** N_{2D}: generated maps remain close to training fields.
- **Larger** N_{2D}: UNet-64 / 128 / 256 move toward novel samples; the wider UNet-256 transition is shifted to larger data.

Calibration results



Main result: error bars show 16–84% seed scatter at fixed $\boldsymbol{\theta}_{\rm ask}$. In the high-data regime they bracket the requested Ω_m; under memorization they can be small but centered on the wrong value. Slopes (0.26 vs 0.79) are secondary summaries; Ω_m is shown because the held-out VGG encoder is strongest there ($R^2 = 0.91$).

Physical fidelity



Across a single run, generated power spectra move toward the real reference from early to final checkpoints. Early checkpoints suppress power over many k bins; by the final checkpoint the generated $P(k)$ follows the real shape much more closely.

Takeaways

1. **Memorization is detectable.** Nearest-neighbor and reproducibility tests locate a data-size transition in CAMELS HI maps.
2. **Conditioning is recoverable.** High-data generated fields track $\boldsymbol{\theta}_{\rm ask}$ much better than memorization-regime fields.
3. **Fidelity requires generalization.** Useful emulators must be novel, statistically faithful, and calibrated before inference use.